

Article

Random Drift Particle Swarm Optimization Algorithm Based on Riemannian Manifolds

Yeerjiang Halimu ^{1,2} , Min Shan ^{3,4}, Chao Zhou ^{3,4} and Jun Sun ^{3,4,*} ¹ School of Internet of Things Engineering, Jiangnan University, Wuxi 214122, China; yej@xjau.edu.cn² School of Computer and Information Engineering, Xinjiang Agricultural University, Urumqi 830052, China³ School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi 214122, China; 7233115001@stu.jiangnan.edu.cn (M.S.); 7223115020@stu.jiangnan.edu.cn (C.Z.)⁴ Key Laboratory of Advanced Process Control for Light Industry, (Ministry of Education), Wuxi 214122, China

* Correspondence: junsun@jiangnan.edu.cn

Abstract

In this paper, we propose the Manifold Random Drift Particle Swarm Optimization (MRDPSO) algorithm for matrix optimization on smooth manifolds. Conventional swarm intelligence methods generally converge prematurely in constrained domains. To mitigate this issue, we introduce the swarm intelligence methods to the manifold and a Random Drift mechanism that regulates the search process. Using Riemannian geometry, our framework treats constrained problems as unconstrained ones on the manifold, which preserves the intrinsic geometric structure of the data. Particles are initialized on the manifold, while updates are performed in tangent spaces. Since geodesic calculations are computationally expensive, we use an inverse retraction as a faster alternative to standard logarithmic mapping. Numerical experiments on Grassmann, Stiefel, and Oblique manifolds show that MRDPSO achieves higher accuracy and superior convergence stability compared to recent state-of-the-art manifold-adapted heuristics, namely IISSO and MSSO.

Keywords: riemannian manifold; random drift particle swarm optimization; tangent space; retraction; inverse retraction; matrix optimization

MSC: 58C05

1. Introduction

Riemannian manifold optimization deals with problems of the form:

$$\min f(\mathbf{X}) \text{ s.t. } \mathbf{X} \in \mathcal{M} \quad (1)$$

where $\mathcal{M}$ denotes a Riemannian manifold and $f : \mathcal{M} \rightarrow \mathbb{R}$ is a real-valued objective function. The framework applies to matrix optimization on smooth spaces, including tasks constrained by nonlinear manifolds or orthogonality [1]. Since Riemannian geometry offers an intrinsic representation independent of local coordinates, it can manage nonlinearities that single-coordinate systems struggle to model. These features explain the use of Riemannian optimization in aerospace, wireless communications, automatic control, computer vision, and machine learning. For instance, Laus et al. [2] developed a non-local means-based algorithm for image denoising on manifolds, and Hintermüller et al. [3] applied Riemannian frameworks to video separation.



Academic Editor: Simeon Reich

Received: 2 March 2026

Revised: 26 March 2026

Accepted: 27 March 2026

Published: 30 March 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

In many engineering cases, variables are found on non-Euclidean spaces like curves or surfaces, instead of flat vector spaces. Applying traditional methods usually requires vectorization or projection, which can destroy the data's geometric integrity. To address these issues, Riemannian optimization operates directly on the manifold itself. This technique converts the constrained problem into an unconstrained one, preserving the geometric properties required for accuracy. As a result, it is used to address problems that were previously difficult to solve [4–6].

Luenberger's [7] early work established the foundation for Riemannian optimization by using geodesics—the shortest paths on a manifold—to guide the search. In this framework, line searches are performed strictly along these paths to ensure the objective function decreases. However, computing exact geodesics is often computationally prohibitive and mathematically complex for many manifolds.

To solve this, Absil et al. [1] suggested using retractions. Instead of calculating exact geodesics and parallel transport, their method uses approximations via retractions and vector transports. This makes the algorithms much more flexible while still maintaining convergence. Subsequently, Riemannian optimization has been integrated with methods including line search [8,9], Newton's method [10], conjugate gradient [11,12], and trust-region algorithms [13,14].

These gradient methods converge quickly at a local level. However, they often fail to reach the global optimum in non-convex problems. This limitation highlights the need for swarm intelligence methods adapted to manifolds, which are better suited for navigating complex, non-convex search spaces. Among them, standard Particle Swarm Optimization (PSO) is widely adopted due to its versatility, demonstrated in applications ranging from complex industrial control systems [15] to large-scale optimization. Yet it frequently suffers from rapid loss of population diversity. To address this, Random Drift Particle Swarm Optimization (RDPSO) [16,17] emerges as a robust alternative. Inspired by free electron motion in an external electric field, RDPSO explicitly decomposes the velocity update into directed drift for local search and thermal motion for global exploration. This mechanism regulates the search process, maintaining a proper mix of exploration and exploitation to prevent early convergence. As a result, RDPSO performs reliably on complex global search tasks even without gradient information.

Following the progress of swarm intelligence, recent developments have introduced advanced paradigms, including the Improved Immune Shark Smell Optimization (IISSO) [18] and the Modified Shark Smell Optimization (MSSO) [19], which currently establish new benchmarks in Euclidean spaces. However, the direct application of these Euclidean heuristics to Riemannian manifolds reveals a fundamental geometric incompatibility. Their heuristic operators—such as Euclidean distance-based immune selection or linear gradient estimation—suffer from structural distortions when directly projected onto curved topologies. Consequently, a critical gap persists in designing algorithms that are inherently compatible with non-Euclidean geometries. To address this gap, the proposed Manifold Random Drift Particle Swarm Optimization (MRDPSO) explicitly abandons heuristic modifications in favor of strict tangent-space dynamics. By formulating physical random drift strictly within the local geometry through inverse retractions, MRDPSO eliminates the geometric distortion associated with conventional Euclidean algorithms.

In this paper, we extend the classical RDPSO algorithm to manifold-valued domains. Our proposed method, Manifold Random Drift Particle Swarm Optimization (MRDPSO), is built on a retraction framework. The main contributions of this paper are:

1. **Riemannian RDPSO Dynamics:** We extend the RDPSO algorithm to Riemannian manifolds by calculating the drift and thermal motions in the tangent space. This matches the underlying geometry while keeping the algorithm's search capabilities intact.

2. **Efficient Inverse Retractions:** The computational expense of exact logarithmic maps is well known. We bypass these geodesic evaluations by adopting inverse retractions. This first-order approximation directly accelerates the velocity update phase. At the same time, it captures sufficient geometric structure to ensure the algorithm converges.
3. **Global Search on Non-Convex Problems:** Standard manifold swarm algorithms often stop too early. We fix this by adding random drift to the Riemannian setup. Because of this change, MRDPSO can handle non-convex spaces that usually trap gradient methods.

To evaluate the performance of MRDPSO, we selected three representative matrix manifold problems as our benchmarks. Specifically, we started with the Stiefel manifold for problems requiring strict orthogonality. Then, we used the Grassmann manifold to test subspace invariance. We also chose the Oblique manifold, which requires column normality but not orthogonality. Since these manifolds cover distinct topological features, they provide a good test for our algorithm's flexibility. (Definitions are in Section 2). These tests allow us to show that MRDPSO works well across a range of matrix optimization tasks.

This is how the paper is structured. In Section 2, we review the fundamental Riemannian manifold concepts, and in Section 3, we describe the proposed MRDPSO framework in addition to the classical RDPSO. Our experimental results and analysis are covered in Section 4, and our conclusions and future directions are outlined in Section 5.

2. Fundamental Knowledge

This section introduces the Riemannian geometry concepts needed for MRDPSO.

2.1. Riemannian Manifolds and Tangent Spaces

A Riemannian manifold $(\mathcal{M}, g)$ is a smooth manifold $\mathcal{M}$ equipped with a Riemannian metric g , which induces a smoothly varying inner product $g_{\mathbf{X}}(\cdot, \cdot)$ on each tangent space $T_{\mathbf{X}}\mathcal{M}$ at $\mathbf{X} \in \mathcal{M}$ [1]. Formally, $T_{\mathbf{X}}\mathcal{M}$ is defined as the set of equivalence classes of smooth curves passing through $\mathbf{X}$, as illustrated in Figure 1, characterizing the local linear approximation of the manifold [1].

$$T_{\mathbf{X}}\mathcal{M} = \mathcal{C}_{\mathbf{X}} / \sim = \{[c] : c \in \mathcal{C}_{\mathbf{X}}\} \quad (2)$$

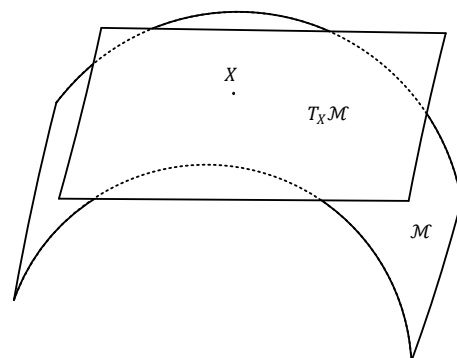


Figure 1. Tangent Space on a Manifold. The dashed line represents a smooth curve passing through point $\mathbf{X}$ on the manifold $\mathcal{M}$.

The disjoint union of all tangent spaces forms the tangent bundle, denoted by $T\mathcal{M} := \cup_{\mathbf{X} \in \mathcal{M}} T_{\mathbf{X}}\mathcal{M}$. In this study, we treat $\mathcal{M}$ as a submanifold embedded in a Euclidean space $\mathbb{R}^{n \times p}$. Consequently, $T_{\mathbf{X}}\mathcal{M}$ is a subspace of the ambient space, and any vector $\mathbf{V} \in \mathbb{R}^{n \times p}$ can be mapped onto the tangent space via an orthogonal projection operator $P_{\mathbf{X}} : \mathbb{R}^{n \times p} \rightarrow T_{\mathbf{X}}\mathcal{M}$ [1].

A tangent vector is strictly bound to its base point. Moving vectors for comparison involves parallel transport $\mathcal{P}_{Y \leftarrow X}$ [1]. Since this exact transport is path-dependent and expensive to compute, we replace it with retractions and inverse retractions. This substitution makes the PSO dynamics computationally feasible while preserving the underlying geometry.

2.2. Retractions and Inverse Retractions

The core operations of MRDPSO involve transitioning between the nonlinear manifold and the linear tangent space. A retraction $R_X : T_X \mathcal{M} \rightarrow \mathcal{M}$ is a smooth mapping that “moves” a point from the tangent space back to the manifold while satisfying $R_X(0_X) = x$ and $DR_X(0_X) = \text{Id}_{T_X \mathcal{M}}$ [1,20]. As illustrated in Figure 2, the retraction transforms a tangent vector $\xi \in T_X \mathcal{M}$ into a unique point on the manifold, effectively providing a local mapping from a linear space to a curved geometry [1].

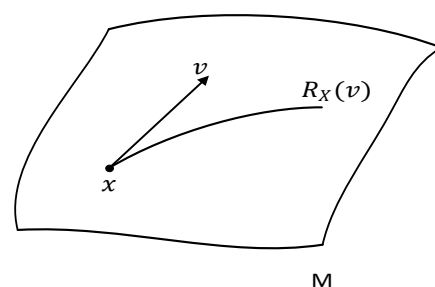


Figure 2. Illustration of a retraction mapping from the tangent space $T_X \mathcal{M}$ to the manifold $\mathcal{M}$. The arrow represents a tangent vector v in the tangent space $T_X \mathcal{M}$, and the curve illustrates its mapping $R_X(v)$ onto the manifold.

Particle velocities depend on pulling a target point $Y \in \mathcal{M}$ back to the tangent space through the inverse retraction $R_X^{-1} : \mathcal{M} \rightarrow T_X \mathcal{M}$. The resulting vector ξ satisfies $R_X \xi = y$. This first-order approximation sidesteps the computational expense of the logarithmic map [1]. It serves as the basis for the drift and stochastic terms in the velocity update.

2.3. Characterization of Matrix Manifolds

We summarize the properties of the three matrix manifolds used in our experiments. For each manifold, we define its structure and the corresponding projection operator P_X required for tangent space operations.

- (1) **Stiefel Manifold:** The Stiefel manifold $St(p, n)$ [21,22] consists of $n \times p$ matrices with orthonormal columns: $St(p, n) = \{X \in \mathbb{R}^{n \times p} : X^\top X = I_p\}$. The tangent space at X is $T_X St = \{U \in \mathbb{R}^{n \times p} : X^\top U + U^\top X = 0\}$. The orthogonal projection of a matrix V onto $T_X St$ is

$$P_X(V) = V - X \text{sym}(X^\top V) \quad (3)$$

where $\text{sym}(A) = \frac{1}{2}(A + A^\top)$.

- (2) **Grassmann Manifold:** $Gr(p, n)$ [23] represents the set of all p -dimensional subspaces in $\mathbb{R}^n$. A tangent vector is uniquely represented by its horizontal lift U satisfying $X^\top U = 0$. The projection onto the horizontal space is:

$$P_X(V) = (I_n - XX^\top)V \quad (4)$$

- (3) Oblique Manifold: $Ob(n, p)$ [24] is the set of matrices with unit-norm columns: $Ob(n, p) = \{\mathbf{X} \in \mathbb{R}^{n \times p} : \text{diag}(\mathbf{X}^\top \mathbf{X}) = \mathbf{1}_p\}$. Its tangent space is $T_{\mathbf{X}}Ob = \{\mathbf{U} \in \mathbb{R}^{n \times p} : \text{diag}(\mathbf{X}^\top \mathbf{U}) = \mathbf{0}\}$. The projection operator is:

$$P_{\mathbf{X}}(\mathbf{V}) = \mathbf{V} - \mathbf{X} \text{ddiag}(\mathbf{X}^\top \mathbf{V}) \quad (5)$$

where $\text{ddiag}(\cdot)$ extracts the diagonal elements and sets the off-diagonal elements to zero.

3. Algorithm Introduction

3.1. Random Drift Particle Swarm Optimization

Random Drift Particle Swarm Optimization is a swarm intelligence algorithm inspired by the free electron model of metal conductors in an external electric field [16]. It establishes the particle velocity equation by simulating the drift and thermal motions of electrons, which guides the swarm toward positions of minimum potential energy. To achieve local search, the drift velocity component directs particles toward both their personal and global best positions [17]. In contrast, the stochastic velocity component derived from thermal motion represents the particles' global search capability. In this section, we formulate the dynamics of RDPSO in a d -dimensional Euclidean space $\mathbb{R}^d$. Consider a swarm of M particles. At the t -th iteration, the position and velocity of the i -th particle are represented by the vectors $\mathbf{x}_i(t) = [x_{i,1}(t), \dots, x_{i,d}(t)]^\top$ and $\mathbf{v}_i(t) = [v_{i,1}(t), \dots, v_{i,d}(t)]^\top$, respectively. The personal best position recorded by the i -th particle is defined as $\mathbf{P}_{best,i}(t)$, while the global best position associated with the swarm is denoted as $\mathbf{G}_{best}(t)$. For a minimization problem $f(\mathbf{x})$, the personal best position $\mathbf{P}_{best,i}(t)$ is updated as:

$$\mathbf{P}_{best,i}(t) = \begin{cases} \mathbf{x}_i(t), & \text{if } f(\mathbf{x}_i(t)) < f(\mathbf{P}_{best,i}(t-1)) \\ \mathbf{P}_{best,i}(t-1), & \text{if } f(\mathbf{x}_i(t)) \geq f(\mathbf{P}_{best,i}(t-1)) \end{cases} \quad (6)$$

The global best position $\mathbf{G}_{best}(t)$ is selected such that $f(\mathbf{G}_{best}(t)) = \min_{1 \leq i \leq M} f(\mathbf{P}_{best,i}(t))$.

The convergence behavior of the swarm is governed by a local attractor, denoted as $\mathbf{P}_{local,i}(t)$ [17]. This attractor represents a stochastic weighted centroid, defined for the j -th dimension ($j = 1, \dots, d$) as:

$$\mathbf{P}_{local,i,j}(t) = \phi_{i,j}(t) \cdot \mathbf{P}_{best,i,j}(t) + (1 - \phi_{i,j}(t)) \cdot \mathbf{G}_{best,j}(t) \quad (7)$$

where $\phi_{i,j}(t)$ is a random variable uniformly distributed in $(0, 1)$.

The kinematic updates in RDPSO are decomposed into a thermal motion component $\mathbf{v}_{R,i,j}$ and a drift motion component $\mathbf{v}_{D,i,j}$, formulated as follows [16]:

$$\mathbf{v}_{R,i,j}(t) = \alpha \cdot |\mathbf{C}_j(t) - \mathbf{x}_{i,j}(t)| \cdot \phi_{i,j}(t) \quad (8)$$

$$\mathbf{v}_{D,i,j}(t) = \beta \cdot (\mathbf{P}_{local,i,j}(t) - \mathbf{x}_{i,j}(t)) \quad (9)$$

Here, $\mathbf{C}(t) = \frac{1}{M} \sum_{i=1}^M \mathbf{P}_{best,i}(t)$ represents the mean best position [16] of the swarm, α is the contraction-expansion coefficient, β acts as the drift coefficient, and $\phi_i(t) \sim N(0, 1)$ is a random variable following a standard normal distribution. The updated position is obtained by:

$$\mathbf{x}_{i,j}(t+1) = \mathbf{x}_{i,j}(t) + \mathbf{v}_{R,i,j}(t) + \mathbf{v}_{D,i,j}(t) \quad (10)$$

3.2. Random Drift Particle Swarm Optimization on Riemannian Manifolds

3.2.1. Stochastic Velocity Component $\mathbf{V}_R$

The stochastic velocity component, $\mathbf{V}_{R,i}(t) \in T_{\mathbf{X}_i}\mathcal{M}$, enables the global exploration of the swarm. Modeled on the thermal motion of free electrons, the scaling coefficient α (which determines the magnitude of the stochastic step) follows a Maxwell–Boltzmann distribution [16], where the probability density function (PDF) is given by:

$$f(\alpha) = \sqrt{\frac{2}{\pi}} \frac{\alpha^2}{\sigma^3} \exp\left(-\frac{\alpha^2}{2\sigma^2}\right) \quad (11)$$

where σ is a parameter related to the thermal intensity.

In Euclidean space, this physical model justifies the use of an adaptive stochastic update:

$$\mathbf{v}_{R,i}(t) = \alpha \cdot |\mathbf{C}(t) - \mathbf{x}_i(t)| \cdot \varphi_i(t) \quad (12)$$

where $\varphi_i(t) \sim N(0, 1)$. Since linear vector operations are undefined on nonlinear manifolds, we map the particle dynamics into the tangent space $T_{\mathbf{X}_i}\mathcal{M}$ [25] using the orthogonal projection operator $\text{Proj}_{\mathbf{X}_i}$ as follows [20,22]:

$$\mathbf{V}_{R,i}(t) = \alpha \cdot \|\mathbf{R}_{\mathbf{X}_i}^{-1}(\mathbf{C}(t))\|_F \cdot \text{Proj}_{\mathbf{X}_i}(\Phi_i(t)) \quad (13)$$

where $\mathbf{R}_{\mathbf{X}_i}^{-1}$ denotes the inverse retraction mapping at the current position $\mathbf{X}_i$, and $\|\cdot\|_F$ is the Frobenius norm, which approximates the geodesic distance to the mean best position $\mathbf{C}(t)$ to scale the exploration step. Furthermore, $\Phi_i(t) \in \mathbb{R}^{n \times p}$ is a random perturbation matrix whose entries follow an independent and identically distributed (i.i.d.) standard normal distribution $N(0, 1)$. Unlike a scalar random variable, this matrix-valued noise ensures that the stochastic exploration covers the full dimensionality of the tangent space rather than being strictly restricted to a 1D trajectory [22].

In the standard RDPSO, the mean best position $\mathbf{C}(t)$ is simply the arithmetic average of all personal best positions. On a Riemannian manifold, however, this becomes the Riemannian center of mass (or Fréchet mean) [1,14], denoted as $\mathbf{C}_{R_center}$:

$$\mathbf{C}_{R_center}(t) = \arg \min_{\mathbf{X} \in \mathcal{M}} \sum_{i=1}^M \text{dist}^2(\mathbf{X}, \mathbf{P}_{best,i}(t)) \quad (14)$$

where $\text{dist}(\cdot, \cdot)$ represents the geodesic distance on the manifold $\mathcal{M}$ and $\mathbf{P}_{best,i}(t)$ denotes the personal best position matrix.

However, calculating the Fréchet mean iteratively is computationally prohibitive ($O(N \cdot K_{iter})$) and may lead to premature convergence by pulling the swarm toward a single deterministic center too early [1]. To address this, we propose a “Stochastic Attractor Strategy.” Instead of computing the exact geometric center, we use a random sample from the current distribution of personal bests to serve as a dynamic guide. This strategy interprets the mean best position not as a fixed point, but as a stochastic estimator:

$$\mathbf{C}_{R_center}(t) \approx \mathbf{P}_{best,random}(t) \in \{\mathbf{P}_{best,1}(t), \mathbf{P}_{best,2}(t), \dots, \mathbf{P}_{best,M}(t)\} \quad (15)$$

Here, the approximation symbol “ $\approx$ ” signifies that the selected $\mathbf{P}_{best,random}$ acts as a single-sample Monte Carlo estimator of the true swarm center. While this introduces variance compared to the exact Fréchet mean, this variance acts as beneficial “exploration noise,” similar to the mechanism in Stochastic Gradient Descent (SGD). This stochasticity keeps the swarm from stalling at local optima across non-convex manifolds.

The stochastic velocity $\mathbf{V}_{R,i}$ on the manifold is thus expressed as shown in Equation (13). This formulation strikes a critical balance: it drastically reduces computational complexity to $O(1)$ while enhancing the global search capability through diversity preservation [17,25].

3.2.2. Drift Velocity Component $\mathbf{V}_D$

The drift velocity component, $v_{D,i}$, drives local exploitation. In Euclidean space, it is given by [16]:

$$v_{D,i}(t) = \beta \cdot (\mathbf{P}_{local,i}(t) - \mathbf{x}_i(t)) \quad (16)$$

where $\mathbf{P}_{local,i}$ is the local attraction point. As particles converge, their positions, personal bests, and the global best all approach this attractor. To extend this mechanism to Riemannian manifolds, we define the local attractor $\mathbf{P}_{local,i}$ by selecting a point on the geodesic path connecting $\mathbf{P}_{best,i}(t)$ and $\mathbf{G}_{best}(t)$ [1]:

$$\mathbf{P}_{local,i}(t) = \gamma_i(t, \phi) \quad (17)$$

where γ_i is the geodesic curve such that $\gamma_i(0) = \mathbf{P}_{best,i}$ and $\gamma_i(1) = \mathbf{G}_{best}$, with $\phi \in (0, 1)$ being a random variable. To maintain computational efficiency, this point on the manifold can be effectively approximated using the retraction operator [1,20]. Mapping this attractor into the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ yields the drift velocity [25]:

$$\mathbf{V}_{D,i}(t) = \beta \cdot R_{\mathbf{x}_i}^{-1}(\mathbf{P}_{local,i}(t)) \quad (18)$$

The drift velocity $\mathbf{V}_{D,i}$ guides the particle toward the local attractor, acting as an adaptive target that blends individual and collective knowledge. By using the inverse retraction $R_{\mathbf{x}_i}^{-1}$, we map this target to a tangent vector that points in a descent direction along the manifold's geometry [1]. Regarding numerical robustness, although inverse retractions are formally local diffeomorphisms, the specific algebraic forms employed for the Stiefel, Grassmann, and Oblique manifolds remain well-defined almost everywhere [23,24]. Therefore, $\mathbf{V}_{D,i}$ acts as a robust guidance field in initial iterations, regardless of the distance to the attractor. This approach eliminates the numerical instability common in exact logarithmic maps [1]. Applying stochasticity to $\mathbf{P}_{local,i}$ disrupts early convergence to suboptimal points. The random perturbations expand the search radius, aiding the swarm in passing through saddle points while preserving the intrinsic manifold structure [1].

Furthermore, to ensure numerical stability and prevent the particle from exceeding the injectivity radius during early extensive exploration [17,20], we implemented a strict velocity clamping mechanism. Before updating the position, the total tangent velocity $\mathbf{V}_i(t) = \mathbf{V}_{R,i}(t) + \mathbf{V}_{D,i}(t)$, it is bounded by a maximum threshold $\mathbf{V}_{max}$, such that if $\mathbf{V}_{max}$, it is scaled to $\mathbf{V}_{max} \cdot \frac{\mathbf{V}_i(t)}{\|\mathbf{V}_i(t)\|_F}$. This geometrically preserves the search direction while ensuring safe and valid retractions.

3.2.3. Inverse Retraction

In Riemannian optimization, the logarithmic map $\log_{\mathbf{x}}\mathbf{Y}$ [1,21] typically serves to pull points from the manifold $\mathcal{M}$ back to the tangent space $T_{\mathbf{x}}\mathcal{M}$. However, exact evaluations of this map demand solving nonlinear matrix equations or tracking expensive geodesics. These operations scale poorly, creating severe computational overhead and numerical issues when the dimensionality increases.

To overcome these limitations, we use the inverse retraction $R_{\mathbf{x}}^{-1}$ —the local inverse of a retraction $R_{\mathbf{x}}$ —as a key strategy in MRDPSO [1]. Compared to geodesic mappings, the inverse retraction offers a more computationally efficient approximation of displacement on the manifold. It is worth noting that while geodesic paths are geometrically natural,

their high computational cost does not necessarily translate to superior optimization performance in swarm-based heuristics.

- Theoretical Properties

1. Local Diffeomorphism and Injectivity: Standard geometric theory [1,20] indicates that a retraction functions as a local diffeomorphism in the vicinity of the tangent space's zero vector 0_X . For any given point X , one can identify a neighborhood around 0_X where the retraction is a smooth bijection. The supremum of the radius of such a neighborhood is termed the retraction injectivity radius. For any point Y within this radius from X on the manifold, the inverse retraction $R_X^{-1}(Y)$ is uniquely and well-defined. To prevent the tangent velocities from exceeding this critical radius during early exploration, the velocity clamping strategy implemented in MRDPSO explicitly bounds the step sizes, ensuring that the retraction maps remain mathematically valid and numerically stable.
2. First-Order Agreement with the Logarithmic Map: We apply the inverse retraction primarily to obtain a first-order approximation of the logarithmic map [1,26]. Provided two points X and Y are sufficiently close, we have:

$$R_X^{-1}(Y) = \log_X(Y) + O(\text{dist}(X, Y)^2) \quad (19)$$

where $\text{dist}(X, Y)$ denotes the Riemannian distance. By this property, the inverse retraction captures the true geodesic direction up to the first order. This bypasses the heavy computational overhead of the exact logarithmic map.

3. Numerical Stability: Logarithmic maps are often prone to numerical errors, especially on matrix manifolds involving matrix exponentials. In contrast, inverse retractions built on stable linear algebra decompositions, such as QR or SVD, offer greater robustness [20]. These features are particularly important for MRDPSO, which requires frequent velocity updates across the population.
- Retraction and Inverse Retraction on Specific Manifolds

1. Grassmann Manifold: The retraction is defined via the singular value decomposition (SVD) [22]:

$$R_X(\xi) = UV^\top \quad (20)$$

where U and V denote the left and right singular matrices of $X + \xi$, respectively. The corresponding inverse retraction is given by [22]:

$$R_X^{-1}(Y) = Y(X^\top Y)^{-1} - X \quad (21)$$

2. Stiefel Manifold: The retraction is defined using the Q-factor of the QR decomposition [1,20]:

$$R_X(\xi) = \text{qf}(X + \xi) \quad (22)$$

The inverse retraction is expressed as [20]:

3. Oblique Manifold: The retraction is performed via column-wise normalization [23]:

$$R_X^{-1}(Y) = YU^{-1} - X \quad (23)$$

where $\mathbf{U}$ is the upper triangular factor derived from the decomposition, $\mathbf{U} = \text{rf}(\mathbf{X} + \boldsymbol{\xi})$.

$$R_{\mathbf{X}}(\boldsymbol{\xi}) = \left[\frac{(\mathbf{X} + \boldsymbol{\xi})_{:,1}}{\|(\mathbf{X} + \boldsymbol{\xi})_{:,1}\|_2}, \dots, \frac{(\mathbf{X} + \boldsymbol{\xi})_{:,p}}{\|(\mathbf{X} + \boldsymbol{\xi})_{:,p}\|_2} \right] \quad (24)$$

The inverse retraction is defined as [23]:

$$R_{\mathbf{X}}^{-1}(\mathbf{Y}) = \mathbf{Y} \cdot \text{diag}(\mathbf{X}^{\top} \mathbf{Y})^{-1} - \mathbf{X} \quad (25)$$

where $\text{diag}(\mathbf{X}^{\top} \mathbf{Y})^{-1}$ represents the diagonal matrix formed by the inverse of the inner products of corresponding columns.

In our implementation, the exact logarithmic map is replaced by the inverse retraction. This alternative provides a reliable first-order approximation of the geodesic while avoiding the numerical instability inherent in exact mappings. Since MRDPSO requires continuous velocity updates in the tangent space, such computational robustness is strictly required to keep the swarm on the manifold.

3.2.4. The Workflow of the MRDPSO

At the t -th iteration, the algorithmic procedure of MRDPSO is executed through the following six steps:

1. **Selection of Mean Best Position:** From the current set of personal best positions $\{\mathbf{P}_{best,1}(t), \dots, \mathbf{P}_{best,M}(t)\}$, a candidate is randomly selected to serve as the approximate mean best position $\mathbf{C}(t)$.
2. **Particle Update Initialization:** Begin the loop for each particle i ($i = 1, \dots, M$) to perform the subsequent geometric updates in the tangent space.
3. **Stochastic Velocity Update:** Calculate the stochastic velocity component $\mathbf{V}_{R,i}(t)$ according to Equation (13). This step generates an isotropic tangent vector in the tangent space $T_{\mathbf{X}_i}\mathcal{M}$ directed from the current position $\mathbf{X}_i(t)$ toward the selected mean best position $\mathbf{C}(t)$, ensuring multidimensional exploration across the manifold [22,25].
4. **Drift Velocity Update:** Calculate the drift velocity component $\mathbf{V}_{D,i}(t)$ according to Equation (18). First, a local attraction point $\mathbf{P}_{local,i}(t)$ is determined by stochastically selecting a point on the geodesic connecting the personal best $\mathbf{P}_{best,i}(t)$ and the global best $\mathbf{G}_{best}(t)$. Subsequently, the inverse retraction $R_{\mathbf{X}_i}^{-1}$ is used to generate a tangent vector in $T_{\mathbf{X}_i}\mathcal{M}$ pointing from $\mathbf{X}_i(t)$ toward $\mathbf{P}_{local,i}(t)$.
5. **Velocity Combination:** Compute the resultant velocity vector $\mathbf{V}_i(t)$ in the tangent space $T_{\mathbf{X}_i}\mathcal{M}$ by linearly combining the stochastic velocity $\mathbf{V}_{R,i}(t)$ and the drift velocity $\mathbf{V}_{D,i}(t)$:

$$\mathbf{V}_i(t) = \mathbf{V}_{R,i}(t) + \mathbf{V}_{D,i}(t) \quad (26)$$

To ensure numerical stability and respect the injectivity radius, the resultant velocity is bounded by $\mathbf{V}_{max}$ (as detailed in Section 3.2.2). Notably, unlike classical PSO, Equation (26) omits the inertia component $\mathbf{V}_i(t-1)$. In Riemannian geometry, adding vectors from distinct tangent spaces necessitates computationally prohibitive vector transport [1]. Furthermore, the foundational RDPSO model [17] assumes memoryless thermal fluctuations rather than classical momentum. Consequently, MRDPSO relies exclusively on the multidimensional isotropic thermal velocity $\mathbf{V}_{R,i}(t)$ to sustain structural exploration and escape local optima, bypassing the geometric overhead of historical velocity tracking.

6. **Position Update via Retraction:** Map the resultant tangent vector $\mathbf{V}_i(t)$ back onto the manifold using the retraction operator $R_{\mathbf{X}_i}$, thereby obtaining the updated particle position:

$$\mathbf{X}_i(t+1) = R_{\mathbf{X}_i}(\mathbf{V}_i(t)) \quad (27)$$

The Pseudocode of the MRDPSO is shown in Algorithm 1.

Algorithm 1. Random Drift Particle Swarm Optimization On Riemannian Manifold (MRDPSO)

Input: $\mathcal{M}$: Riemannian manifold; $f(\cdot)$: Objective function; $R_{\mathbf{X}}$: Retraction; $R_{\mathbf{X}}^{-1}$: Inverse retraction; α, β : Algorithmic coefficients; $\mathbf{V}_{max}$: Maximum velocity threshold; M : Population size; T_{max} : Maximum iterations.

Output: $\mathbf{G}_{best}$: Found optimal solution.

```

1: Set  $t \leftarrow 0$ 
2: Initialize population  $\{\mathbf{X}_i(0)\}_{i=1}^M$  randomly on  $\mathcal{M}$ 
3: Initialize personal bests:  $\mathbf{P}_{best,i} \leftarrow \mathbf{X}_i(0)$  for  $i = 1, \dots, M$ 
4: Evaluate fitness:  $f(\mathbf{X}_i(0))$  for  $i = 1, \dots, M$ 
5: Initialize global best  $\mathbf{G}_{best}$  such that  $f(\mathbf{G}_{best}) = \min_i f(\mathbf{P}_{best,i})$ 
6: while  $t < T_{max}$  do
7:   // Step 1: Selection of Mean Best Position
8:   Select  $\mathbf{C}(t)$  randomly from the set  $\{\mathbf{P}_{best,1}, \dots, \mathbf{P}_{best,M}\}$ 
9:   for  $i = 1$  to  $M$  do
10:    // Step 4: Determine local attraction point  $\mathbf{P}_{local,i}$ 
11:    Select a random point  $\mathbf{P}_{local,i}$  on the geodesic between  $\mathbf{P}_{best,i}$  and  $\mathbf{G}_{best}$ 
12:    // Step 3 & 4: Map dynamics to tangent space  $T_{\mathbf{X}_i}\mathcal{M}$ 
13:     $\mathbf{V}_{D,i} \leftarrow \beta \cdot R_{\mathbf{X}_i}^{-1}(\mathbf{P}_{local,i})$  / Drift component
14:     $\mathbf{V}_{R,i} \leftarrow \alpha \cdot \|R_{\mathbf{X}_i}^{-1}(\mathbf{C}(t))\|_F \cdot \text{Proj}_{\mathbf{X}_i}(\Phi_i)$  // Stochastic component,  $\Phi_i \sim \mathcal{N}(0, 1)^{n \times p}$ 
15:    // Step 5: Combine velocities
16:     $\mathbf{V}_i \leftarrow \mathbf{V}_{D,i} + \mathbf{V}_{R,i}$ 
17:    if  $\|\mathbf{V}_i\|_F > \mathbf{V}_{max}$  then
18:       $\mathbf{V}_i \leftarrow \mathbf{V}_{max} \cdot \frac{\mathbf{V}_i}{\|\mathbf{V}_i\|_F}$ 
19:    end if
20:    // Step 6: Position update via retraction
21:     $\mathbf{X}_i(t+1) \leftarrow R_{\mathbf{X}_i}(\mathbf{V}_i)$ 
22:  end for
23:  // Update Personal and Global Bests
24:  for  $i = 1$  to  $M$  do
25:    if  $f(\mathbf{X}_i(t+1)) < f(\mathbf{P}_{best,i})$  then
26:       $\mathbf{P}_{best,i} \leftarrow \mathbf{X}_i(t+1)$ 
27:    end if
28:  end for
29:  if  $\min_i f(\mathbf{P}_{best,i}) < f(\mathbf{G}_{best})$  then
30:    Update  $\mathbf{G}_{best}$  with the best particle among  $\{\mathbf{P}_{best,i}\}$ 
31:  end if
32:   $t \leftarrow t + 1$ 
33: end while
34: return  $\mathbf{G}_{best}$ 

```

4. Experiments

4.1. Comparative Experiments

To evaluate the performance of our MRDPSO algorithm, we chose two state-of-the-art metaheuristic variants as baselines: the Improved Immune Shark Smell Optimization (IISO) [18] and the Modified Shark Smell Optimization (MSSO) [19]. These algorithms, published between 2024 and 2025, represent recent advancements in swarm intelligence and allow for a more rigorous assessment against current optimization standards.

To ensure a valid comparison, the baseline algorithms were implemented as manifold-constrained variants. This was achieved by mapping their update dynamics into the tangent space and employing the same retraction and inverse retraction operators as MRDPSO [1,27]. We used the default parameters for these baselines as specified in their original publications [18,19], setting the convergence tolerance to 10^{-8} .

To ensure a fair comparison, all algorithms are subject to the same termination conditions. The maximum number of function evaluations (MaxFEs) is primarily determined by the manifold dimensions as $n \times p \times 5000$. This high evaluation budget is intentionally allocated to ensure that the stochastic search can adequately navigate the complex, non-convex landscapes inherent to high-dimensional matrix manifolds.

Each experiment was repeated 30 times independently. Numerical results for all test cases are reported as Mean $\pm$ Standard Deviation to characterize both convergence precision and algorithmic stability. Since the results from stochastic algorithms do not necessarily follow a normal distribution, we conducted the non-parametric Wilcoxon rank-sum test with a significance level of 0.05 to verify if the performance differences between MRDPSO and the baselines are statistically significant. Furthermore, an early-exit rule was implemented: the search halts if the relative improvement in the best objective value stays below 10^{-5} between consecutive steps [28]. These settings help balance high-precision convergence with computational speed.

4.2. Parameter Settings

Particle dynamics in MRDPSO are defined by thermal and drift components, which represent global search and local refinement. Following the formulation in Section 3.2, the drift term $\mathbf{V}_{D,i}$ facilitates local convergence by shifting particle positions toward $\mathbf{G}_{best}$ and the individual $\mathbf{P}_{best,i}$. This ensures that the particle's current position $\mathbf{X}_i$ and its $\mathbf{P}_{best,i}$ continuously approach the swarm's optimal region. Conversely, the thermal motion $\mathbf{V}_{R,i}$ induces stochastic fluctuations, allowing particles to reach positions far from the current attractors. This component is essential for enhancing global search and preventing premature convergence on complex, non-convex manifolds. Accordingly, the balance between these two components dictates the overall convergence behavior.

Two coefficients, the stochastic factor α and the drift velocity factor β , regulate the dynamics of the MRDPSO. We base the selection of these parameters on the theoretical foundations of RDPSO [16] and adapt them to meet the geometric constraints of Riemannian manifolds. According to the convergence analysis established by Sun et al. [16], the coefficients must satisfy the following conditions to ensure that particle trajectories remain probabilistically bounded:

$$0 < \alpha < 1, \quad 0 < \beta < 2 \quad (28)$$

Within this theoretical framework, α scales the stochastic component to control the exploration intensity, while β determines the rate of convergence toward the local attractor $\mathbf{P}_{local,i}$.

We applied a grid search on the Stiefel manifold to identify parameters within the theoretical range. According to Figure 3, the best fitness (-288.5) was recorded at $\alpha = 0.5$ and $\beta = 1.0$. We found that $\alpha > 0.7$ destabilizes the search, whereas $\alpha < 0.3$ results

in local minima. Notably, although $\beta < 2$ is theoretically sound, settings above 1.4 lead to overshooting and numerical errors during retraction on curved spaces [1,20]. These empirical findings confirm that our selected configuration provides the ideal balance between exploration and exploitation. Although these parameters were optimized using the Stiefel manifold, they generalize effectively to the Grassmann and Oblique manifolds. This generalization is justified because the velocity updates are performed in the local tangent space—which is intrinsically Euclidean—while the chosen values ensure that step sizes remain within the injectivity radii [1] of the respective retractions, thus maintaining numerical stability across different manifold topologies.

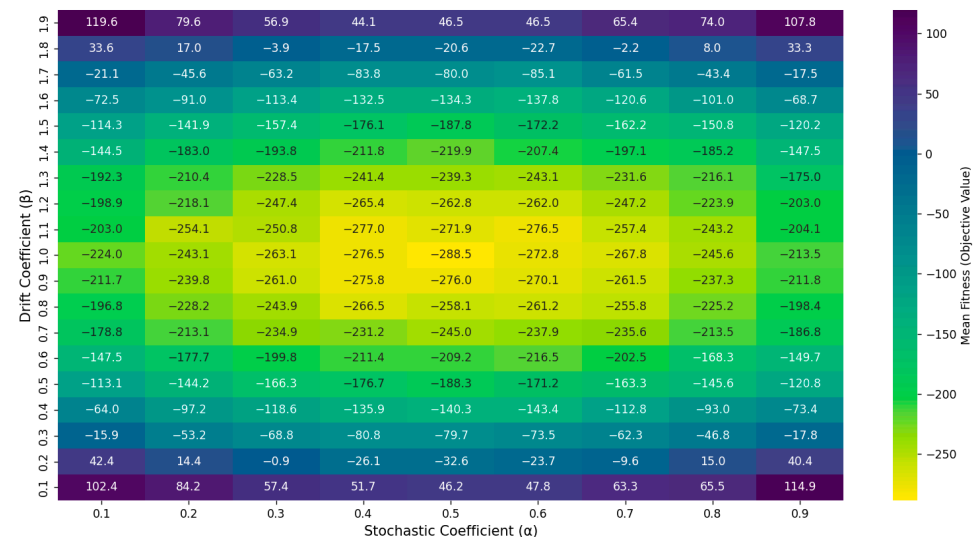


Figure 3. Parameter sensitivity analysis of the stochastic coefficient (α) and drift coefficient (β) on the Stiefel manifold. The heatmap illustrates the mean objective values (Fitness) over 20 independent runs; the yellow region corresponds to the optimal configuration ($\alpha = 0.5$, $\beta = 1.0$) with the minimum fitness of -288.5 .

Consequently, we selected a moderate stochastic intensity and a stable drift strength for our experiments:

$$\alpha = 0.5, \quad \beta = 1.0 \quad (29)$$

These settings yield stable convergence and maintain global search across various manifolds. We fixed the population size at $M = 50$ to limit computational cost. At the same time, this size provides enough diversity to sample the manifold topology adequately at each iteration t .

4.3. Experimental Results

4.3.1. Secant Dimensionality Reduction

The Secant Dimensionality Reduction (SDR) problem [26,29] is utilized as our first benchmark to evaluate the performance of MRDPSO on the Grassmann manifold $Gr(n, p)$. The core of the SDR task is to find a subspace that preserves the structure of a dataset by maximizing the capture of secant directions [26]. In our implementation, the secant set Σ is constructed as follows:

$$\Sigma = \left\{ \frac{\mathbf{x}_{:,i} - \mathbf{x}_{:,j}}{\|\mathbf{x}_{:,i} - \mathbf{x}_{:,j}\|_2} : 1 \leq i < j \leq N \right\} \quad (30)$$

where $\mathbf{x}_{:,i}$ and $\mathbf{x}_{:,j}$ are data points from the input set. The optimization task is formulated as a max-min problem on the Grassmannian [29]:

$$\max_{\mathbf{X} \in Gr(n,p)} \min_{\mathbf{s} \in \Sigma} \|\mathbf{X}^T \mathbf{s}\|_2^2 \quad (31)$$

In our tests, we varied the ambient dimension $n \in \{30, 50, 100, 150, 200\}$ and the subspace dimension $p \in \{3, 7, 9, 11\}$. For each (n, p) configuration, 30 independent sample groups were generated, and each group was tested over 10 independent runs to ensure the reliability of the results. Table 1 reports the resulting performance in terms of the mean and Standard Deviation (Std) of the final objective values.

As illustrated in Figure 4, the convergence curves show that MRDPSO maintains a faster initial descent compared to IISSO [18] and MSSO [19] algorithms. Consistent with our algorithm motivation, the thermal motion component $\mathbf{V}_{R,i}$ induces stochastic fluctuations that prevent the particles from being trapped in local optima [16], which is particularly evident on the curved geometry of $Gr(n, p)$. The baseline algorithms, especially MSSO [19], frequently exhibit stagnation during the search process. In contrast, MRDPSO maintains search diversity and converges to a more precise solution.

Table 1 reports the numerical outcomes. Across various n, p dimensions, MRDPSO records the highest mean fitness. The Standard Deviation (Std) metrics reflect its stability, showing a narrower data spread than both IISSO and MSSO. For the difficult max-min objective, the inverse retraction mechanism sustains search diversity alongside local convergence [25]. This setup yields robust solutions without adding heavy computational overhead. The results of the Wilcoxon rank-sum test ($+/\approx/-$), shown at the bottom of Table 1, confirm that MRDPSO significantly outperforms the two state-of-the-art baselines in all 20 test cases.

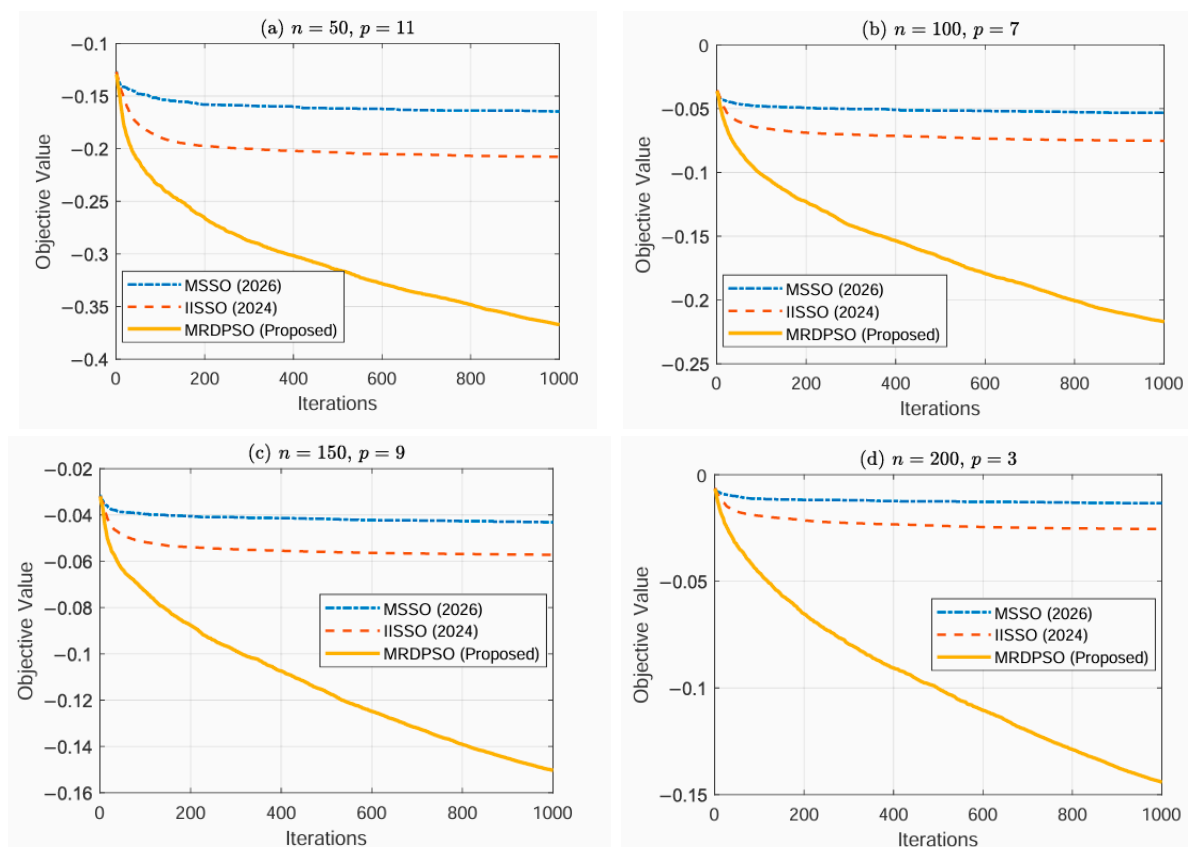


Figure 4. Convergence curves for Secant Dimensionality Reduction on the Grassmann manifold.

Table 1. Numerical performance (Mean $\pm$ Std) for SDR on the Grassmann manifold. Note: Data are expressed as Mean $\pm$ Std. For these minimization tasks, the best value per row is highlighted by a dark background, and the second-best by a light gray background. The “+ / $\approx$ / −” row summarizes the number of wins, ties, and losses according to the Wilcoxon rank-sum test.

n	p	MRDPSO (Proposed)	IISSO (2024)	MSSO (2025)
30	3	$-3.64 \times 10^{-1} \pm 2.3 \times 10^{-2}$	$-1.81 \times 10^{-1} \pm 3.1 \times 10^{-2}$	$-9.84 \times 10^{-2} \pm 6.0 \times 10^{-3}$
30	7	$-4.94 \times 10^{-1} \pm 1.9 \times 10^{-2}$	$-2.54 \times 10^{-1} \pm 2.1 \times 10^{-2}$	$-1.88 \times 10^{-1} \pm 9.5 \times 10^{-3}$
30	9	$-5.29 \times 10^{-1} \pm 2.0 \times 10^{-2}$	$-3.02 \times 10^{-1} \pm 2.3 \times 10^{-2}$	$-2.36 \times 10^{-1} \pm 7.7 \times 10^{-3}$
30	11	$-5.69 \times 10^{-1} \pm 1.6 \times 10^{-2}$	$-3.65 \times 10^{-1} \pm 2.2 \times 10^{-2}$	$-2.91 \times 10^{-1} \pm 8.1 \times 10^{-3}$
50	3	$-2.74 \times 10^{-1} \pm 1.5 \times 10^{-2}$	$-9.88 \times 10^{-2} \pm 1.5 \times 10^{-2}$	$-5.46 \times 10^{-2} \pm 5.0 \times 10^{-3}$
50	7	$-3.42 \times 10^{-1} \pm 1.4 \times 10^{-2}$	$-1.45 \times 10^{-1} \pm 1.4 \times 10^{-2}$	$-1.08 \times 10^{-1} \pm 3.8 \times 10^{-3}$
50	9	$-3.50 \times 10^{-1} \pm 1.6 \times 10^{-2}$	$-1.77 \times 10^{-1} \pm 1.0 \times 10^{-2}$	$-1.35 \times 10^{-1} \pm 6.2 \times 10^{-3}$
50	11	$-3.67 \times 10^{-1} \pm 1.6 \times 10^{-2}$	$-2.08 \times 10^{-1} \pm 1.3 \times 10^{-2}$	$-1.65 \times 10^{-1} \pm 4.7 \times 10^{-3}$
100	3	$-2.38 \times 10^{-1} \pm 1.7 \times 10^{-2}$	$-5.51 \times 10^{-2} \pm 9.8 \times 10^{-3}$	$-2.69 \times 10^{-2} \pm 2.1 \times 10^{-3}$
100	7	$-2.17 \times 10^{-1} \pm 1.2 \times 10^{-2}$	$-7.52 \times 10^{-2} \pm 7.0 \times 10^{-3}$	$-5.32 \times 10^{-2} \pm 3.0 \times 10^{-3}$
100	9	$-2.14 \times 10^{-1} \pm 1.1 \times 10^{-2}$	$-9.08 \times 10^{-2} \pm 5.7 \times 10^{-3}$	$-6.57 \times 10^{-2} \pm 2.6 \times 10^{-3}$
100	11	$-2.18 \times 10^{-1} \pm 1.0 \times 10^{-2}$	$-1.04 \times 10^{-1} \pm 7.2 \times 10^{-3}$	$-7.98 \times 10^{-2} \pm 2.8 \times 10^{-3}$
150	3	$-1.89 \times 10^{-1} \pm 1.3 \times 10^{-2}$	$-3.90 \times 10^{-2} \pm 5.2 \times 10^{-3}$	$-1.77 \times 10^{-2} \pm 1.7 \times 10^{-3}$
150	7	$-1.61 \times 10^{-1} \pm 9.1 \times 10^{-3}$	$-5.11 \times 10^{-2} \pm 4.4 \times 10^{-3}$	$-3.57 \times 10^{-2} \pm 2.0 \times 10^{-3}$
150	9	$-1.50 \times 10^{-1} \pm 7.1 \times 10^{-3}$	$-5.72 \times 10^{-2} \pm 5.8 \times 10^{-3}$	$-4.32 \times 10^{-2} \pm 2.0 \times 10^{-3}$
150	11	$-1.49 \times 10^{-1} \pm 7.0 \times 10^{-3}$	$-6.90 \times 10^{-2} \pm 4.9 \times 10^{-3}$	$-5.22 \times 10^{-2} \pm 1.6 \times 10^{-3}$
200	3	$-1.44 \times 10^{-1} \pm 7.9 \times 10^{-3}$	$-2.55 \times 10^{-2} \pm 3.5 \times 10^{-3}$	$-1.34 \times 10^{-2} \pm 1.3 \times 10^{-3}$
200	7	$-1.20 \times 10^{-1} \pm 9.0 \times 10^{-3}$	$-3.81 \times 10^{-2} \pm 3.7 \times 10^{-3}$	$-2.55 \times 10^{-2} \pm 9.5 \times 10^{-4}$
200	9	$-1.16 \times 10^{-1} \pm 6.5 \times 10^{-3}$	$-4.47 \times 10^{-2} \pm 3.9 \times 10^{-3}$	$-3.20 \times 10^{-2} \pm 1.0 \times 10^{-3}$
200	11	$-1.11 \times 10^{-1} \pm 6.4 \times 10^{-3}$	$-5.00 \times 10^{-2} \pm 2.8 \times 10^{-3}$	$-3.86 \times 10^{-2} \pm 1.1 \times 10^{-3}$
+ / $\approx$ / −		20/0/0		20/0/0

4.3.2. Joint Diagonalization Problem

This experiment evaluates the MRDPSO algorithm on the Joint Diagonalization of symmetric matrices [23], a benchmark problem also explored in [30]. The objective is to find an orthogonal matrix $\mathbf{X}$ on the Stiefel manifold $St(n, p)$ that minimizes the squared off-diagonal elements of the transformed matrices. The cost function is defined as:

$$\min_{\mathbf{X} \in St(n, p)} f(\mathbf{X}) = \sum_{k=1}^K \left(\|\mathbf{X}^\top \mathbf{A}_k \mathbf{X}\|_F^2 - \|\text{diag}(\mathbf{X}^\top \mathbf{A}_k \mathbf{X})\|_F^2 \right) \quad (32)$$

In this formulation, the matrix $\mathbf{X}$ satisfies the orthogonality constraint $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, and each matrix $\mathbf{A}_k$ is symmetric. Specifically, K represents the total number of symmetric matrices to be diagonalized simultaneously.

The experiments cover ambient dimensions $n \in \{30, 50, 100, 150\}$ and subspace dimensions $p \in \{3, 5, 9, 11\}$, forming 20 distinct test cases. For statistical consistency, we executed 30 independent runs per case. Table 2 reports the final objective values using the Mean and Standard Deviation (Std).

The convergence curves in Figure 5 show that MRDPSO yields a steadier descent and lower final objective values than IISSO [18] and MSSO [19]. As intended by the design, the thermal motion $\mathbf{V}_{R,i}(t)$ introduces stochastic noise that keeps the swarm from stalling at local minima. This randomness is highly effective across the curved topology of $St(n, p)$ [20]. While the baseline algorithms, particularly MSSO [19], often encounter stagnation in higher dimensions, MRDPSO sustains search diversity and converges to a more precise solution.

Based on the data in Table 2, MRDPSO achieves the lowest objective values in every one of the 20 test cases. The Standard Deviation (Std) values further indicate high consistency compared to baseline methods. Facing difficult parameter settings, the proposed approach

maintains search diversity and limits execution time. The synergy of drift velocity terms and inverse retractions on the manifold drives this stable output [20,25]. According to the Wilcoxon rank-sum test (+/≈/−), MRDPSO demonstrates a statistically significant improvement (20/0/0) over both modern baselines.

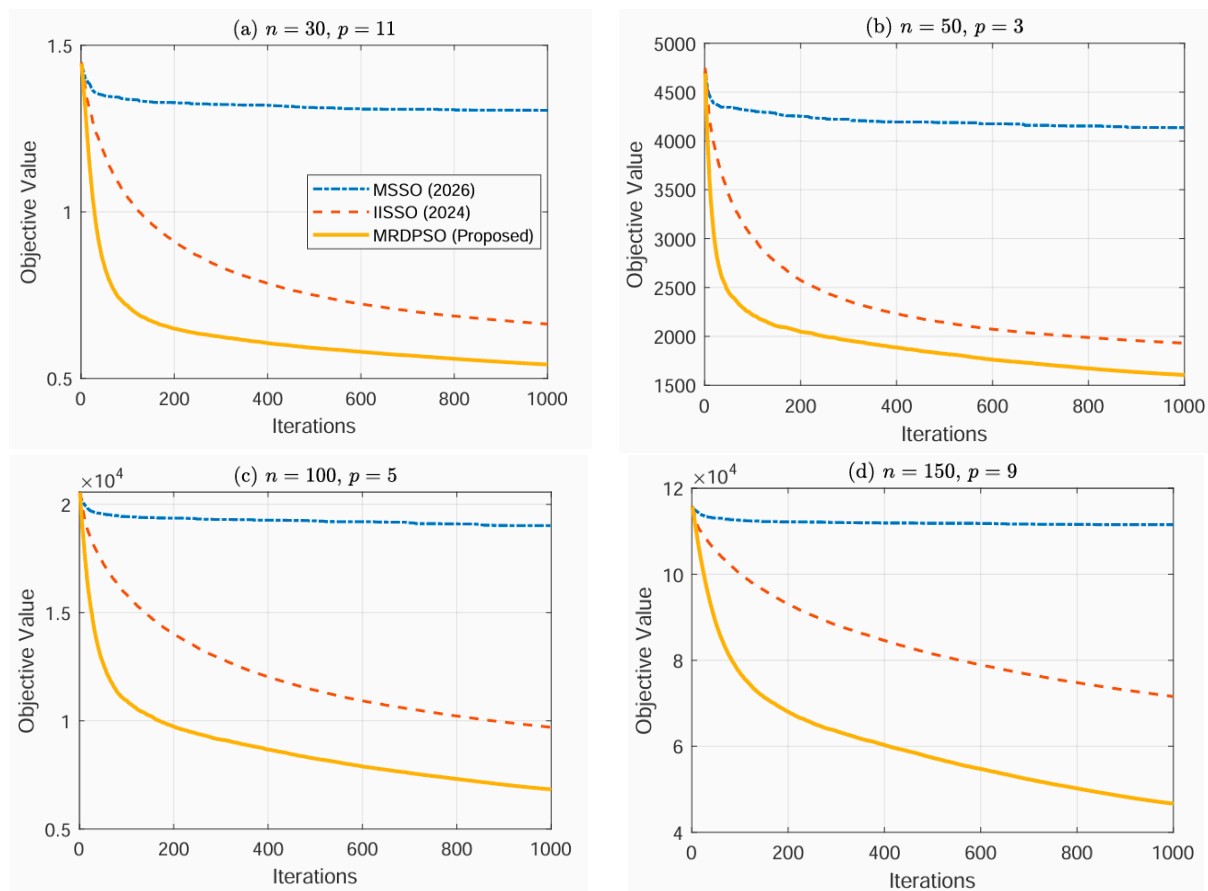


Figure 5. Convergence curves for selected experiments on the Joint Diagonalization problem.

Table 2. Numerical performance (Mean $\pm$ Std) for the Joint Diagonalization problem on the Stiefel manifold. Note: Data are expressed as Mean $\pm$ Std. For these minimization tasks, the best value per row is highlighted by a dark background, and the second-best by a light gray background. The “+ / $\approx$ / −” row summarizes the number of wins, ties, and losses according to the Wilcoxon rank-sum test.

n	p	MRDPSO (Proposed)	IISSO (2024)	MSSO (2025)
30	3	$1.41 \times 10^3 \pm 1.0 \times 10^2$	$1.61 \times 10^3 \pm 1.1 \times 10^2$	$3.42 \times 10^3 \pm 7.4 \times 10^1$
30	5	$9.65 \times 10^3 \pm 2.3 \times 10^2$	$1.08 \times 10^4 \pm 3.7 \times 10^2$	$1.90 \times 10^4 \pm 2.3 \times 10^2$
30	9	$5.97 \times 10^4 \pm 1.4 \times 10^3$	$6.62 \times 10^4 \pm 1.2 \times 10^3$	$1.08 \times 10^5 \pm 9.1 \times 10^2$
30	11	$5.42 \times 10^4 \pm 1.8 \times 10^3$	$6.63 \times 10^4 \pm 2.1 \times 10^3$	$1.30 \times 10^5 \pm 1.4 \times 10^3$
50	3	$1.60 \times 10^3 \pm 8.6 \times 10^1$	$1.93 \times 10^3 \pm 1.5 \times 10^2$	$4.14 \times 10^3 \pm 7.6 \times 10^1$
50	5	$7.45 \times 10^3 \pm 2.4 \times 10^2$	$9.04 \times 10^3 \pm 3.4 \times 10^2$	$1.80 \times 10^4 \pm 1.8 \times 10^2$
50	9	$5.19 \times 10^4 \pm 6.6 \times 10^2$	$6.31 \times 10^4 \pm 1.1 \times 10^3$	$1.08 \times 10^5 \pm 9.9 \times 10^2$
50	11	$8.29 \times 10^4 \pm 1.3 \times 10^3$	$1.06 \times 10^5 \pm 1.9 \times 10^3$	$1.80 \times 10^5 \pm 1.7 \times 10^3$
100	3	$2.17 \times 10^3 \pm 9.8 \times 10^1$	$2.66 \times 10^3 \pm 9.6 \times 10^1$	$5.09 \times 10^3 \pm 4.0 \times 10^1$
100	5	$6.83 \times 10^3 \pm 2.5 \times 10^2$	$9.70 \times 10^3 \pm 2.7 \times 10^2$	$1.90 \times 10^4 \pm 2.0 \times 10^2$
100	9	$4.69 \times 10^4 \pm 6.0 \times 10^2$	$6.71 \times 10^4 \pm 8.6 \times 10^2$	$1.10 \times 10^5 \pm 6.1 \times 10^2$
100	11	$7.04 \times 10^4 \pm 1.1 \times 10^3$	$1.08 \times 10^5 \pm 1.5 \times 10^3$	$1.80 \times 10^5 \pm 1.1 \times 10^3$

Table 2. Cont.

n	p	MRDPSO (Proposed)	IISSO (2024)	MSSO (2025)
150	3	$1.92 \times 10^3 \pm 1.2 \times 10^2$	$2.67 \times 10^3 \pm 1.1 \times 10^2$	$5.09 \times 10^3 \pm 4.1 \times 10^1$
150	5	$8.88 \times 10^3 \pm 2.0 \times 10^2$	$1.27 \times 10^4 \pm 2.6 \times 10^2$	$2.16 \times 10^4 \pm 1.2 \times 10^2$
150	9	$4.67 \times 10^4 \pm 6.6 \times 10^2$	$7.16 \times 10^4 \pm 8.7 \times 10^2$	$1.12 \times 10^5 \pm 4.4 \times 10^2$
150	11	$6.98 \times 10^4 \pm 1.0 \times 10^3$	$1.15 \times 10^5 \pm 1.4 \times 10^3$	$1.82 \times 10^5 \pm 7.3 \times 10^2$
200	3	$2.26 \times 10^3 \pm 7.7 \times 10^1$	$3.05 \times 10^3 \pm 7.1 \times 10^1$	$5.38 \times 10^3 \pm 4.0 \times 10^1$
200	5	$8.36 \times 10^3 \pm 1.8 \times 10^2$	$1.28 \times 10^4 \pm 1.7 \times 10^2$	$2.14 \times 10^4 \pm 1.2 \times 10^2$
200	9	$4.76 \times 10^4 \pm 6.7 \times 10^2$	$7.54 \times 10^4 \pm 9.0 \times 10^2$	$1.12 \times 10^5 \pm 5.6 \times 10^2$
200	11	$7.09 \times 10^4 \pm 8.5 \times 10^2$	$1.21 \times 10^5 \pm 8.1 \times 10^2$	$1.82 \times 10^5 \pm 1.2 \times 10^3$
+/ $\approx$ /−		20/0/0		20/0/0

4.3.3. Max-Cut Problem

In this subsection, the performance of the MRDPSO algorithm is evaluated on the Max-Cut problem [29]. For an undirected weighted graph, the Max-Cut problem seeks a partition of vertices into two disjoint sets such that the cumulative weight of the edges spanning the partition is maximized. This combinatorial problem can be relaxed and reformulated as a continuous minimization problem on the Oblique manifold $Ob(n, p)$ [29], expressed as:

$$\min_{\mathbf{X} \in Ob(n, p)} f(\mathbf{X}) = \text{tr}(\mathbf{X}^\top \mathbf{L} \mathbf{X}) \quad (33)$$

In this formulation:

- $\mathbf{X}$ is the decision matrix on the Oblique manifold, where each column $\mathbf{x}_i$ is constrained to the unit sphere ($\mathbf{x}_i^\top \mathbf{x}_i = 1$). While the standard Max-Cut problem typically utilizes $p = 1$, the multi-column case represents a higher-rank relaxation or multi-way partition.
- $\mathbf{L}$ denotes the graph Laplacian matrix, defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is the diagonal degree matrix with entries $d_{ii} = \sum_j a_{ij}$, and $\mathbf{A}$ is the symmetric adjacency matrix containing the edge weights.
- $\text{tr}(\cdot)$ denotes the matrix trace operator. Minimizing the negative trace (or the positive trace of the Laplacian form) corresponds to maximizing the quadratic form associated with the cut weight [29].

We selected the number of vertices $n \in \{30, 50, 100, 150\}$ and the rank $p \in \{3, 5, 9, 11\}$, yielding 20 distinct test settings. For each setting, 30 independent runs were performed. The results are reported using the Mean and Standard Deviation (Std) of the objective values, as shown in Table 3.

As illustrated in Figure 6, the convergence curves show that MRDPSO maintains a steadier descent and reaches lower final objective values compared to IISSO [18] and MSSO [19]. Consistent with our algorithm motivation, the thermal motion component $\mathbf{V}_{R,i}(t)$ induces stochastic fluctuations that help particles escape local traps [16]. While the baseline algorithms frequently show early stagnation on the Oblique manifold, MRDPSO continues to explore the search space, eventually reaching a higher-precision solution.

Based on Table 3, MRDPSO returns the strongest results across the 20 test cases. It reaches smaller objective minimums and, matching the low Standard Deviation (Std) data, exhibits lower initialization sensitivity. This behavior confirms that the inverse retraction and random drift components operate correctly within the Oblique manifold $Ob(n, p)$. The Wilcoxon rank-sum test (+/ $\approx$ /−) confirms that MRDPSO achieves statistically significant superiority (20/0/0) over both contemporary baselines.

Table 3. Numerical performance (Mean $\pm$ Std) for the Max-Cut problem on the Oblique manifold. Note: Data are expressed as Mean $\pm$ Std. For these minimization tasks, the best value per row is highlighted by a dark background, and the second-best by a light gray background. The “+ / $\approx$ / −” row summarizes the number of wins, ties, and losses according to the Wilcoxon rank-sum test.

n	p	MRDPSO (Proposed)	IISO (2024)	MSSO (2025)
30	3	$2.27 \times 10^{-2} \pm 2.5 \times 10^{-3}$	$1.10 \times 10^{00} \pm 6.7 \times 10^{-1}$	$1.09 \times 10^1 \pm 3.1 \times 10^{-1}$
30	5	$4.50 \times 10^{-2} \pm 7.2 \times 10^{-3}$	$3.25 \times 10^{00} \pm 1.2 \times 10^{00}$	$2.04 \times 10^1 \pm 4.3 \times 10^{-1}$
30	9	$2.64 \times 10^{-1} \pm 3.8 \times 10^{-1}$	$1.08 \times 10^1 \pm 1.7 \times 10^{00}$	$4.02 \times 10^1 \pm 5.4 \times 10^{-1}$
30	11	$3.44 \times 10^{-1} \pm 3.2 \times 10^{-1}$	$1.48 \times 10^1 \pm 1.7 \times 10^{00}$	$5.06 \times 10^1 \pm 5.6 \times 10^{-1}$
50	3	$6.84 \times 10^{-2} \pm 6.4 \times 10^{-3}$	$3.32 \times 10^{00} \pm 2.0 \times 10^{00}$	$2.40 \times 10^1 \pm 3.9 \times 10^{-1}$
50	5	$1.40 \times 10^{-1} \pm 9.0 \times 10^{-3}$	$1.00 \times 10^1 \pm 2.9 \times 10^{00}$	$4.26 \times 10^1 \pm 6.5 \times 10^{-1}$
50	9	$5.92 \times 10^{-1} \pm 5.7 \times 10^{-1}$	$2.94 \times 10^1 \pm 4.1 \times 10^{00}$	$8.08 \times 10^1 \pm 5.5 \times 10^{-1}$
50	11	$1.26 \times 10^{00} \pm 1.6 \times 10^{00}$	$4.07 \times 10^1 \pm 2.9 \times 10^{00}$	$1.00 \times 10^2 \pm 6.0 \times 10^{-1}$
100	3	$3.16 \times 10^{-1} \pm 3.3 \times 10^{-2}$	$1.18 \times 10^1 \pm 3.2 \times 10^{00}$	$5.09 \times 10^1 \pm 4.2 \times 10^{-1}$
100	5	$1.15 \times 10^{00} \pm 1.9 \times 10^{00}$	$3.35 \times 10^1 \pm 5.1 \times 10^{00}$	$8.74 \times 10^1 \pm 5.0 \times 10^{-1}$
100	9	$3.10 \times 10^{00} \pm 1.6 \times 10^{00}$	$8.90 \times 10^1 \pm 4.9 \times 10^{00}$	$1.61 \times 10^2 \pm 6.2 \times 10^{-1}$
100	11	$7.83 \times 10^{00} \pm 5.2 \times 10^{00}$	$1.19 \times 10^2 \pm 5.6 \times 10^{00}$	$1.98 \times 10^2 \pm 6.7 \times 10^{-1}$
150	3	$9.44 \times 10^{-1} \pm 9.7 \times 10^{-2}$	$2.78 \times 10^1 \pm 5.8 \times 10^{00}$	$8.35 \times 10^1 \pm 4.2 \times 10^{-1}$
150	5	$2.91 \times 10^{00} \pm 2.0 \times 10^{00}$	$7.37 \times 10^1 \pm 6.9 \times 10^{00}$	$1.42 \times 10^2 \pm 5.0 \times 10^{-1}$
150	9	$1.34 \times 10^1 \pm 7.9 \times 10^{00}$	$1.72 \times 10^2 \pm 7.8 \times 10^{00}$	$2.61 \times 10^2 \pm 8.1 \times 10^{-1}$
150	11	$2.58 \times 10^1 \pm 8.7 \times 10^{00}$	$2.22 \times 10^2 \pm 7.8 \times 10^{00}$	$3.20 \times 10^2 \pm 7.6 \times 10^{-1}$
200	3	$1.95 \times 10^{00} \pm 1.2 \times 10^{-1}$	$4.72 \times 10^1 \pm 7.5 \times 10^{00}$	$1.11 \times 10^2 \pm 5.6 \times 10^{-1}$
200	5	$6.56 \times 10^{00} \pm 2.6 \times 10^{00}$	$1.09 \times 10^2 \pm 5.9 \times 10^{00}$	$1.87 \times 10^2 \pm 5.1 \times 10^{-1}$
200	9	$3.00 \times 10^1 \pm 1.1 \times 10^1$	$2.43 \times 10^2 \pm 8.1 \times 10^{00}$	$3.41 \times 10^2 \pm 4.8 \times 10^{-1}$
200	11	$5.28 \times 10^1 \pm 1.7 \times 10^1$	$3.14 \times 10^2 \pm 8.1 \times 10^{00}$	$4.18 \times 10^2 \pm 1.1 \times 10^{00}$
+ / $\approx$ / −		20/0/0		20/0/0

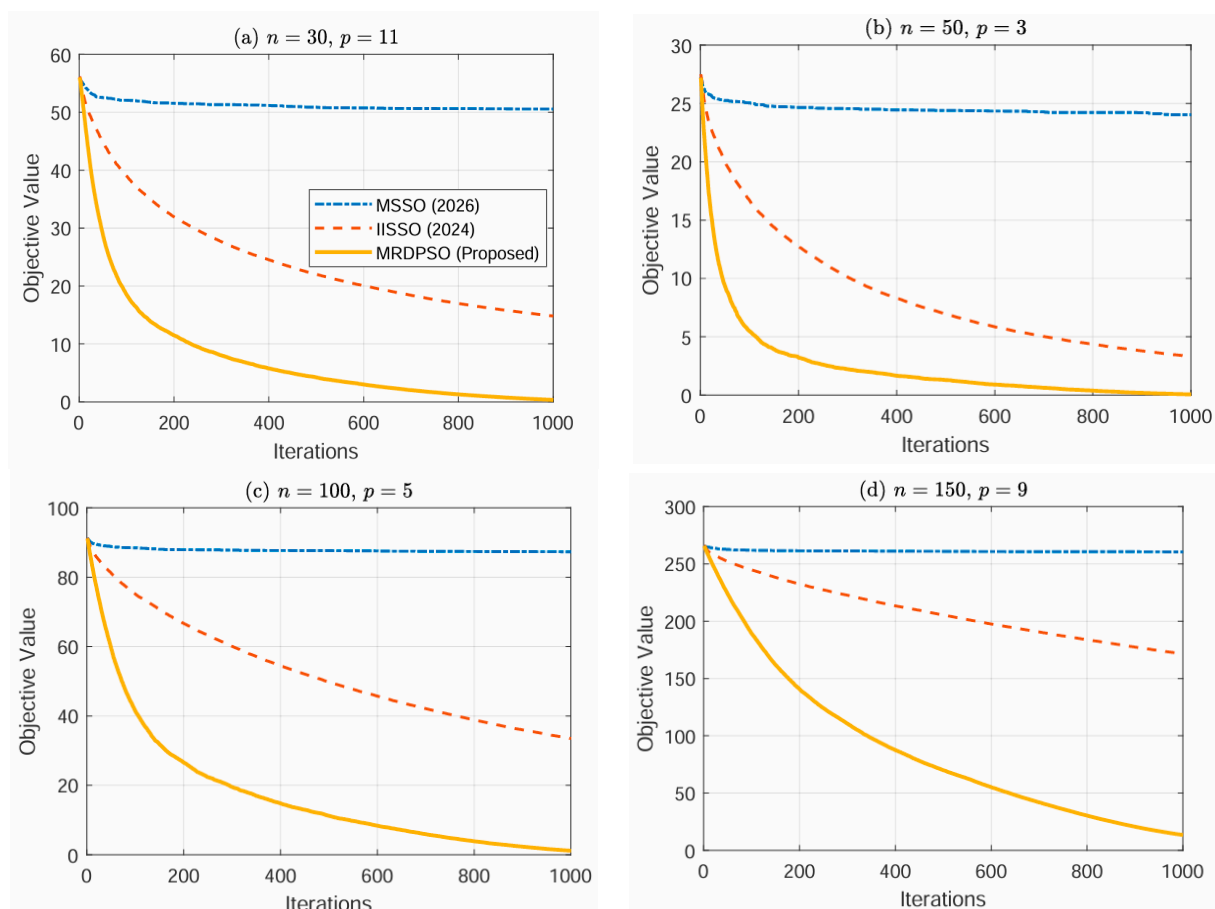


Figure 6. Convergence curves for selected experiments on the Max-Cut problem.

4.3.4. Analysis of Comparative Experimental Results

We applied the Wilcoxon rank-sum test ($p < 0.05$) to measure performance gaps between MRDPSO and the baseline methods. Tables 1–3 document this comparative data for MRDPSO, IISSO [18], and MSSO [19] on the tested manifold problems. Across all tested dimensions and problem settings, MRDPSO consistently attains smaller objective values and shows steadier convergence behavior. The statistical scores of 20/0/0 recorded in all three tables demonstrate that the proposed method achieved significant superiority across all sixty test cases.

Observations show that IISSO and MSSO exhibit relatively fast initial convergence but stall at local minima once swarm diversity drops. This behavior worsens significantly on high-dimensional manifolds [1]. Where the non-convexity is more pronounced [26]. Despite the advanced immune and gradient mechanisms in these baselines, they demonstrate limited robustness when navigating the intrinsic geometric constraints of Riemannian manifolds.

In contrast, MRDPSO delivers consistent results across all benchmarks by integrating drift updates $\mathbf{V}_{D,i}$ with thermal motion $\mathbf{V}_{R,i}$ on the manifold [16]. This combined mechanism prevents stagnation at local minima and sustains necessary population diversity [17]. Tables 1–3 document this stability through narrower Standard Deviation (Std) spreads, indicating that the algorithm depends less on initial states.

Figures 4–6 plot the convergence across the three manifold problems. MRDPSO records a smoother descent, fewer oscillations, and smaller final objective values than both IISSO and MSSO. These data confirm that lifting the RDPSO dynamics into the tangent space $T_{\mathbf{x}}\mathcal{M}$ via inverse retractions sustains the geometric structure and accelerates the search process [20,26].

4.3.5. Ablation Study: Inverse Retraction vs. Logarithmic Map

This ablation study isolates the impact of geometric mapping operators on algorithm performance. Table 4 contrasts the inverse retraction $R_{\mathbf{x}}^{-1}$ against the exact logarithmic map $\log_{\mathbf{x}}$ for the Secant Dimensionality Reduction and Max-Cut tasks [1,26]. Regardless of the dimensional setup, $R_{\mathbf{x}}^{-1}$ records lower objective minimums and smaller standard deviations (Std) compared to the exact logarithmic map.

The difference mainly comes from how the two mappings are computed [1]. Computing the logarithmic map involves explicit geodesic calculations and a large number of matrix operations [1,20]. When the dimension n becomes large, this process can be numerically unstable, particularly when points approach the boundary of the injectivity radius [1]. The inverse retraction instead relies on a simple local approximation and requires less computation [26]. In our experiments, this setting usually shows fewer numerical issues and reaches acceptable solutions with fewer iterations, especially for large or ill-conditioned problems.

Based on these results, replacing the logarithmic map with the inverse retraction allows for faster and more stable optimization [26]. Therefore, it is a suitable option for the MRDPSO method. The Wilcoxon rank-sum test (+/ $\approx$ /–) confirms that the inverse retraction significantly outperforms the logarithmic mapping in all 20 test configurations (20/0/0) for both problem sets.

Table 4. Mean and Standard Deviation (Std) of final results using inverse retraction versus logarithmic mapping. Note: Data are expressed as Mean $\pm$ Std. For these minimization tasks, the best value per row is highlighted by a dark background. The “+ / $\approx$ / −” row summarizes the number of wins, ties, and losses according to the Wilcoxon rank-sum test.

Problem		Secant Dimensionality Reduction		MaxCut	
n	p	Inverse Retraction	Logarithmic Map	Inverse Retraction	Logarithmic Map
30	3	$-3.66 \times 10^{-1} \pm 2.2 \times 10^{-2}$	$-3.64 \times 10^{-1} \pm 2.5 \times 10^{-2}$	$2.28 \times 10^{-2} \pm 3.3 \times 10^{-3}$	$2.24 \times 10^{-2} \pm 2.9 \times 10^{-3}$
30	5	$-4.29 \times 10^{-1} \pm 1.3 \times 10^{-2}$	$-4.29 \times 10^{-1} \pm 2.1 \times 10^{-2}$	$6.98 \times 10^{-2} \pm 1.4 \times 10^{-1}$	$4.44 \times 10^{-2} \pm 6.3 \times 10^{-3}$
30	9	$-5.26 \times 10^{-1} \pm 2.0 \times 10^{-2}$	$-5.32 \times 10^{-1} \pm 1.8 \times 10^{-2}$	$3.43 \times 10^{-1} \pm 4.5 \times 10^{-1}$	$1.79 \times 10^{-1} \pm 1.8 \times 10^{-1}$
30	11	$-5.76 \times 10^{-1} \pm 2.2 \times 10^{-2}$	$-5.70 \times 10^{-1} \pm 1.9 \times 10^{-2}$	$4.05 \times 10^{-1} \pm 4.9 \times 10^{-1}$	$3.39 \times 10^{-1} \pm 3.9 \times 10^{-1}$
50	3	$-2.65 \times 10^{-1} \pm 2.1 \times 10^{-2}$	$-2.68 \times 10^{-1} \pm 2.1 \times 10^{-2}$	$7.21 \times 10^{-2} \pm 6.8 \times 10^{-3}$	$6.97 \times 10^{-2} \pm 7.4 \times 10^{-3}$
50	5	$-3.21 \times 10^{-1} \pm 1.3 \times 10^{-2}$	$-3.19 \times 10^{-1} \pm 2.2 \times 10^{-2}$	$1.43 \times 10^{-1} \pm 1.4 \times 10^{-2}$	$1.67 \times 10^{-1} \pm 1.0 \times 10^{-1}$
50	9	$-3.48 \times 10^{-1} \pm 2.1 \times 10^{-2}$	$-3.53 \times 10^{-1} \pm 1.7 \times 10^{-2}$	$4.33 \times 10^{-1} \pm 2.4 \times 10^{-1}$	$6.03 \times 10^{-1} \pm 6.3 \times 10^{-1}$
50	11	$-3.72 \times 10^{-1} \pm 1.1 \times 10^{-2}$	$-3.74 \times 10^{-1} \pm 1.5 \times 10^{-2}$	$1.10 \times 10^0 \pm 1.0 \times 10^0$	$1.28 \times 10^0 \pm 1.2 \times 10^0$
100	3	$-2.32 \times 10^{-1} \pm 1.8 \times 10^{-2}$	$-2.34 \times 10^{-1} \pm 1.7 \times 10^{-2}$	$3.15 \times 10^{-1} \pm 2.8 \times 10^{-2}$	$3.20 \times 10^{-1} \pm 2.8 \times 10^{-2}$
100	5	$-2.23 \times 10^{-1} \pm 1.1 \times 10^{-2}$	$-2.24 \times 10^{-1} \pm 1.1 \times 10^{-2}$	$1.14 \times 10^0 \pm 2.0 \times 10^0$	$9.62 \times 10^{-1} \pm 1.1 \times 10^0$
100	9	$-2.12 \times 10^{-1} \pm 1.4 \times 10^{-2}$	$-2.13 \times 10^{-1} \pm 1.2 \times 10^{-2}$	$3.56 \times 10^0 \pm 2.5 \times 10^0$	$4.54 \times 10^0 \pm 3.7 \times 10^0$
100	11	$-2.19 \times 10^{-1} \pm 1.1 \times 10^{-2}$	$-2.19 \times 10^{-1} \pm 1.1 \times 10^{-2}$	$7.33 \times 10^0 \pm 4.6 \times 10^0$	$7.82 \times 10^0 \pm 4.6 \times 10^0$
150	3	$-1.89 \times 10^{-1} \pm 1.0 \times 10^{-2}$	$-1.88 \times 10^{-1} \pm 1.4 \times 10^{-2}$	$9.43 \times 10^{-1} \pm 8.9 \times 10^{-2}$	$1.25 \times 10^0 \pm 1.5 \times 10^0$
150	5	$-1.75 \times 10^{-1} \pm 1.0 \times 10^{-2}$	$-1.75 \times 10^{-1} \pm 1.3 \times 10^{-2}$	$2.56 \times 10^0 \pm 5.6 \times 10^{-1}$	$2.40 \times 10^0 \pm 4.6 \times 10^{-1}$
150	9	$-1.49 \times 10^{-1} \pm 9.9 \times 10^{-3}$	$-1.52 \times 10^{-1} \pm 1.0 \times 10^{-2}$	$1.31 \times 10^1 \pm 4.4 \times 10^0$	$1.29 \times 10^1 \pm 6.2 \times 10^0$
150	11	$-1.52 \times 10^{-1} \pm 8.8 \times 10^{-3}$	$-1.49 \times 10^{-1} \pm 6.8 \times 10^{-3}$	$2.32 \times 10^1 \pm 1.1 \times 10^1$	$2.50 \times 10^1 \pm 6.8 \times 10^0$
200	3	$-1.45 \times 10^{-1} \pm 7.6 \times 10^{-3}$	$-1.43 \times 10^{-1} \pm 7.6 \times 10^{-3}$	$1.93 \times 10^0 \pm 1.6 \times 10^{-1}$	$2.05 \times 10^0 \pm 6.5 \times 10^{-1}$
200	5	$-1.35 \times 10^{-1} \pm 9.2 \times 10^{-3}$	$-1.36 \times 10^{-1} \pm 7.2 \times 10^{-3}$	$5.95 \times 10^0 \pm 3.6 \times 10^0$	$7.69 \times 10^0 \pm 6.4 \times 10^0$
200	9	$-1.16 \times 10^{-1} \pm 7.2 \times 10^{-3}$	$-1.16 \times 10^{-1} \pm 6.4 \times 10^{-3}$	$3.36 \times 10^1 \pm 1.1 \times 10^1$	$3.10 \times 10^1 \pm 1.0 \times 10^1$
200	11	$-1.12 \times 10^{-1} \pm 7.3 \times 10^{-3}$	$-1.12 \times 10^{-1} \pm 7.3 \times 10^{-3}$	$5.12 \times 10^1 \pm 1.1 \times 10^1$	$5.40 \times 10^1 \pm 1.7 \times 10^1$
+ / $\approx$ / −		20/0/0		20/0/0	

5. Conclusions

This study develops the Manifold Random Drift Particle Swarm Optimization (MRDPSO) algorithm, moving the RDPSO framework from Euclidean space to Riemannian manifolds. The algorithm computes stochastic (thermal) and drift velocities within each particle’s tangent space using inverse retraction. These components are then merged into a resultant tangent vector, which determines the particle’s new position via a retraction back to the manifold. MRDPSO was tested on three problems: Secant Dimensionality Reduction, Joint Diagonalization, and Max-Cut, defined on different manifolds. During the experiments, it converged faster and produced lower objective values than contemporary baselines such as IISSO [18] and MSSO [19], while showing fewer numerical issues in non-Euclidean settings. Despite these advantages, a current limitation involves the reliance on fixed parameters for the stochastic and drift components, which may require manual tuning to maintain optimal stability on manifolds with exceptionally high curvature.

Future studies will consider adding adaptive parameter control and probability density migration to manifold-based swarm algorithms to improve particle update strategies. Additionally, we aim to extend these algorithmic frameworks to optimization problems on manifolds with more intricate topological structures, such as product manifolds.

Author Contributions: Conceptualization, Y.H. and M.S.; methodology, Y.H. and J.S.; software, C.Z.; formal analysis, Y.H. and M.S.; investigation, C.Z. and M.S.; writing—original draft preparation, Y.H. and M.S.; writing—review and editing, Y.H., M.S. and J.S.; visualization, C.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported in part by the National Natural Science Foundation of China under Grants 62272202 and 61672263.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments: The authors would like to express their sincere thanks to the anonymous referees for their great efforts to improve this paper.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Absil, P.-A.; Mahony, R.; Sepulchre, R. *Optimization Algorithms on Matrix Manifolds*; Princeton University Press: Princeton, NJ, USA, 2008.
2. Laus, F.; Nikolova, M.; Persch, J.; Steidl, G. A Nonlocal Denoising Algorithm for Manifold-Valued Images Using Second Order Statistics. *SIAM J. Imaging Sci.* **2017**, *10*, 416–448. [\[CrossRef\]](#)
3. Hintermüller, M.; Wu, T. Robust Principal Component Pursuit via Inexact Alternating Minimization on Matrix Manifolds. *J. Math. Imaging Vis.* **2015**, *51*, 204–228. [\[CrossRef\]](#)
4. Sarkis, M.; Diepold, K. Camera-Pose Estimation via Projective Newton Optimization on the Manifold. *IEEE Trans. Image Process.* **2012**, *21*, 1729–1741. [\[CrossRef\]](#)
5. Borckmans, P.B.; Selvan, S.E.; Boumal, N.; Absil, P.-A. A Riemannian subgradient algorithm for economic dispatch with valve-point effect. *J. Comput. Appl. Math.* **2014**, *255*, 848–866. [\[CrossRef\]](#)
6. Li, X.; Zhang, L. An abnormal event detection method based on the Riemannian manifold and LSTM network. *Neurocomputing* **2021**, *440*, 144–154. [\[CrossRef\]](#)
7. Luenberger, D.G. The gradient projection method along geodesics. *Manag. Sci.* **1972**, *18*, 620–631. [\[CrossRef\]](#)
8. Meyer, G.; Bonnabel, S.; Sepulchre, R. Regression on Fixed-Rank Positive Semidefinite Matrices: A Riemannian Approach. *J. Mach. Learn. Res.* **2011**, *12*, 593–625.
9. Hosseini, S.; Huang, W.; Yousefpour, R. Line Search Algorithms for Locally Lipschitz Functions on Riemannian Manifolds. *SIAM J. Optim.* **2018**, *28*, 596–619. [\[CrossRef\]](#)
10. Zhao, Z.; Bai, Z.-J.; Jin, X.-Q. A Riemannian Newton Algorithm for Nonlinear Eigenvalue Problems. *SIAM J. Matrix Anal. Appl.* **2015**, *36*, 752–774. [\[CrossRef\]](#)
11. Ring, W.; Wirth, B. Optimization Methods on Riemannian Manifolds and Their Application to Shape Space. *SIAM J. Optim.* **2012**, *22*, 596–627. [\[CrossRef\]](#)
12. Sato, H.; Iwai, T. A new, globally convergent Riemannian conjugate gradient method. *Optimization* **2015**, *64*, 1011–1031. [\[CrossRef\]](#)
13. Huang, W.; Absil, P.-A.; Gallivan, K.A. A Riemannian symmetric rank-one trust-region method. *Math. Program.* **2015**, *150*, 179–216. [\[CrossRef\]](#)
14. Huang, W. Optimization Algorithms on Riemannian Manifolds with Applications. Ph.D. Thesis, Florida State University, Tallahassee, FL, USA, 2013.
15. Tran, N.-H.; Dao, Q.-T.; Nguyen, T.-V.-A.; Bui, N.-T. Trajectory and Parameter Optimization in Robust Tracking Control of a Quadrotor. *IEEE Access* **2025**, *13*, 155215. [\[CrossRef\]](#)
16. Sun, J.; Wu, X.; Palade, V.; Fang, W. Random Drift Particle Swarm Optimization. *Discret. Dyn. Nat. Soc.* **2013**, *2013*, 305219.
17. Sun, J.; Palade, V.; Wu, X.; Fang, W.; Nalepa, J. Multiple Sequence Alignment with Hidden Markov Models Learned by Random Drift Particle Swarm Optimization. *IEEE/ACM Trans. Comput. Biol. Bioinform.* **2014**, *11*, 243–257. [\[CrossRef\]](#)
18. Nguyen, D.-H.; Nguyen, T.-V.-A. Optimized Interval Type-2 Fuzzy Control Using Modified Shark Smell Algorithm for Precision Trajectory Tracking of 8-Cable-Driven Parallel Robots. *Int. J. Fuzzy Syst.* **2026**. [\[CrossRef\]](#)
19. Chen, Y.; Liu, G.; Wang, L.; Xia, B. Automatic Vertical Parking Reference Trajectory Based on Improved Immune Shark Smell Optimization. *Algorithms* **2024**, *17*, 308. [\[CrossRef\]](#)
20. Absil, P.-A.; Malick, J. Projection-like Retractions on Matrix Manifolds. *SIAM J. Optim.* **2012**, *22*, 135–158. [\[CrossRef\]](#)
21. Chen, S.; Garcia, A.; Hong, M.; Shahrampour, S. Decentralized Riemannian Gradient Descent on the Stiefel Manifold. *IEEE Trans. Signal Process.* **2021**, *69*, 1842–1854.
22. Absil, P.-A.; Mahony, R.; Sepulchre, R. Riemannian Geometry of Grassmann Manifolds with a View on Algorithmic Computation. *Acta Appl. Math.* **2004**, *80*, 199–220. [\[CrossRef\]](#)
23. Absil, P.-A.; Gallivan, K.A. Joint Diagonalization on the Oblique Manifold for Independent Component Analysis. In Proceedings of the 2006 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toulouse, France, 14–19 May 2006; Volume 5, pp. 945–948.
24. Manton, J.H. Optimization algorithms exploiting unitary constraints. *IEEE Trans. Signal Process.* **2002**, *50*, 635–650. [\[CrossRef\]](#)
25. Boumal, N.; Mishra, B.; Absil, P.-A.; Sepulchre, R. Manopt, a MATLAB Toolbox for Optimization on Manifolds. *J. Mach. Learn. Res.* **2014**, *15*, 1455–1459.

26. Zhu, X.; Sato, H. Riemannian conjugate gradient methods with inverse retraction. *Comput. Optim. Appl.* **2020**, *77*, 779–810. [[CrossRef](#)]
27. Broomhead, D.S.; Kirby, M.J. Dimensionality Reduction Using Secant-Based Projection Methods: The Induced Dynamics in Projected Systems. *Nonlinear Dyn.* **2005**, *41*, 47–67. [[CrossRef](#)]
28. Polak, E. *Optimization: Algorithms and Consistent Approximations*; Springer: New York, NY, USA, 1997.
29. Journée, M.; Bach, F.; Absil, P.-A.; Sepulchre, R. Low-Rank Optimization on the Cone of Positive Semidefinite Matrices. *SIAM J. Optim.* **2010**, *20*, 2327–2351. [[CrossRef](#)]
30. Absil, P.-A.; Hosseini, S. A Collection of Nonsmooth Riemannian Optimization Problems. In *Nonsmooth Optimization and Its Applications*; Springer: Cham, Switzerland, 2019; pp. 1–31.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.